Indian Institute of Technology Bombay

PH 544 - Midterm 1 **General relativity** **Date 25-02-2026**

Solutions

1. Tangent vector at a typical point on a curve which is parametrized by t is given by [**0.5 pt**]:

$$U^\mu = \frac{dx^\mu}{dt}$$

Components of U^μ are given by [**0.5 pt**]:

$$U^0 = \frac{d}{dt}(\sqrt{3}ct) = \sqrt{3}c$$
$$U^1 = \frac{d}{dt}(ct) = c$$
$$U^2 = \frac{d}{dt}(ct_0\cos(t/t_0)) = -c\sin(t/t_0)$$
$$U^3 = \frac{d}{dt}(ct_0\sin(t/t_0)) = c\cos(t/t_0)$$

In Minkowski spacetime ($\eta_{\mu\nu} = \text{diag}(+1,-1,-1,-1)$), the magnitude squared of the tangent vector is [**0.5 pt**]:

$$|U|^2 = \eta_{\mu\nu}U^\mu U^\nu = (U^0)^2 - (U^1)^2 - (U^2)^2 - (U^3)^2$$
$$|U|^2 = (\sqrt{3}c)^2 - (c)^2 - (-c\sin(t/t_0))^2 - (c\cos(t/t_0))^2 = c^2$$

Since $|U|^2 > 0$, tangent vector any typical point on the curve is **timelike** [**0.5 pt**].

2. (a) In frame S:
Event 1: $x_1 = 0$, $t_1 = 0$
Event 2: $x_2 = (12c)\,\text{m}$, $t_2 = 20\text{s}$
Spacetime interval:

$$ds^2 = c^2dt^2 - dx^2 = 400c^2 - 144c^2 = 256c^2 > 0$$

Since the interval between the two events is timelike, it is possible to find a frame S' in which the two events occur at the same place [**1 pt**].

(b) Let the frame S' move with speed v relative to frame S along the x-axis. In S', the two events occur at the same place $\implies \Delta x' = 0$. Using the Lorentz transformation [**1pt**]

$$\Delta x' = \gamma(\Delta x - v\Delta t)$$
$$v = \frac{\Delta x}{\Delta t} = \frac{12c}{20} = 0.6c$$

3. Covariant derivative of a rank-2 tensor g_{ij} [**0.5 pt**]:

$$\nabla_k g_{ij} = \partial_k g_{ij} - \Gamma^l_{ki} g_{lj} - \Gamma^l_{kj} g_{il}$$

where

$$\Gamma^l_{ki} = \frac{1}{2} g^{lm} (\partial_k g_{im} + \partial_i g_{mk} - \partial_m g_{ki})$$

Since $g_{lj} g^{lm} = \delta^m_j$, 2nd term becomes [**0.5 pt**]:

$$\Gamma^l_{ki} g_{lj} = \frac{1}{2} (\partial_k g_{ij} + \partial_i g_{jk} - \partial_j g_{ki})$$

Similar, 3rd term [**0.5 pt**]:

$$\Gamma^l_{kj} g_{il} = \frac{1}{2} (\partial_k g_{ji} + \partial_j g_{ik} - \partial_i g_{kj})$$

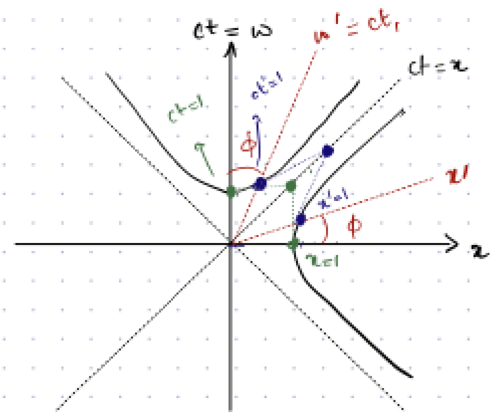
Substituting these back into the covarient derivative formula, we get [**0.5 pt**]

$$\nabla_k g_{ij} = 0$$

4. Spacetime diagram:

   (a) Calibration hyperbolas: **[0.5 pt for hyperbola equations, 0.5 pt for diagram]**

   (b) 1) Green point on $ct = x$ line corresponds to $x = 1, ct = 1$.
   2) Blue point on $ct = x$ line corresponds to $x' = 1, ct' = 1$.
   [1 pt for each point]



$\phi = \tan^{-1}(\beta)$

$\beta = 0.5$

Slopes of $x' = 0$ and $ct' = 0$ lines:

$x' = \gamma(x - \beta w)$ $\quad x' = 0 \Rightarrow x = \beta w \quad$ slope $= \frac{1}{\beta}$

$w' = \gamma(w - \beta x)$ $\quad w' = 0 \Rightarrow w = \beta x \quad$ slope $= \beta$

$$\left.\begin{aligned} w^2 - x^2 &= 1 \\ x^2 - w^2 &= 1 . \end{aligned}\right\} \text{Calibration hyperbolas.}$$

5. The Gaussian curvature is given by [**0.5 pt**]

$$K = \frac{\det(\mathrm{II})}{\det(\mathrm{I})} \tag{1}$$

For the metric [**0.5 pt**]

$$\begin{aligned} ds^2 &= dx^2 + dy^2 + dz^2 \\ &= dx^2 + dy^2 + (d(f(x,y)))^2 \\ &= (1 + f_x^2)dx^2 + (1 + f_y^2)dy^2 + 2f_x f_y dx dy \end{aligned}$$

Hence $\det(\mathrm{I}) = (1 + f_x^2)(1 + f_y^2) - f_x^2 f_y^2$

For the second fundamental form The normal vector is given by [**0.5 pt**]

$$\begin{aligned}\hat{n} &= \frac{\vec{\nabla}\left[f(x,y)-z\right]}{|\vec{\nabla}\left[f(x,y)-z\right]|}\\ &= \frac{1}{\sqrt{f_x^2+f_y^2+1}}\left[f_x, f_y, -1\right]\end{aligned}$$

We also need to second derivative of the position vector $\vec{r}(x,y) = [x, y, f(x,y)]$. We have [**0.5 pt**]

$$\begin{aligned}\vec{r}_{xx} &= [0, 0, f_{xx}]\\ \vec{r}_{xy} &= [0, 0, f_{xy}]\\ \vec{r}_{yy} &= [0, 0, f_{yy}]\end{aligned}$$

The second fundamental form is given by [**0.5 pt**]

$$\begin{aligned}\mathrm{II}(x,y) &= \begin{bmatrix}\vec{r}_{xx}.\hat{n} & \vec{r}_{xy}.\hat{n}\\ \vec{r}_{yx}.\hat{n} & \vec{r}_{yy}.\hat{n}\end{bmatrix}\\ &= -\frac{1}{\sqrt{f_x^2+f_y^2+1}}\begin{bmatrix}f_{xx} & f_{xy}\\ f_{yx} & f_{yy}\end{bmatrix}\end{aligned}$$

Hence the determinant of the second fundamental form is given by

$$\det(\mathrm{II}) = \frac{f_{xx}f_{yy} - f_{xy}^2}{f_x^2+f_y^2+1}$$

Hence [**0.5 pt**]

$$K = \frac{f_{xx}f_{yy} - f_{xy}^2}{(1+f_x^2+f_y^2)^2}$$

6. (a)

$$T_{\mu\nu\sigma\delta} = \lambda(g_{\mu\sigma}g_{\nu\delta} - g_{\mu\delta}g_{\nu\sigma})$$

Antisymmetric in first two indices [**0.5 pt**].

$$\begin{aligned}T_{\nu\mu\sigma\delta} &= \lambda(g_{\nu\sigma}g_{\mu\delta} - g_{\nu\delta}g_{\mu\sigma})\\ T_{\nu\mu\sigma\delta} &= -\lambda(g_{\nu\delta}g_{\mu\sigma} - g_{\nu\sigma}g_{\mu\delta})\\ T_{\nu\mu\sigma\delta} &= -T_{\mu\nu\sigma\delta}\end{aligned}$$

Similarly, antisymmetric in the last two indices [**0.5 pt**].
Symmetric under exchange of pairs ($T_{\mu\nu\sigma\delta} = T_{\sigma\delta\mu\nu}$) is satisfied since the metric is symmetric [**0.5 pt**].
Bianchi identity: All terms will cancel pairwise [**0.5 pt**].

$$T_{\mu\nu\sigma\delta} + T_{\mu\sigma\delta\nu} + T_{\mu\delta\nu\sigma} = 0$$

(b) Ricci tensor [**1 pt**]: $R_{\nu\delta} = g^{\mu\sigma} T_{\mu\nu\sigma\delta}$

$$\begin{aligned} R_{\nu\delta} &= \lambda g^{\mu\sigma}(g_{\mu\sigma}g_{\nu\delta} - g_{\mu\delta}g_{\nu\sigma}) \\ R_{\nu\delta} &= \lambda(4g_{\nu\delta} - g_{\nu\delta}) = 3\lambda g_{\nu\delta} \end{aligned}$$

Ricci scalar [**1 pt**]:

$$\begin{aligned} R = g^{\nu\delta}R_{\nu\delta} &= g^{\nu\delta}(3\lambda g_{\nu\delta}) \\ R &= 3\lambda(4) = 12\lambda \end{aligned}$$

7. (a) The potential at the centre of the Earth due to the Moon's field is [**1 pt**]

$$\Phi(x, y, z) = -\frac{GM_m}{\sqrt{x^2 + y^2 + z^2}}$$

Hence [**0.5 pt + 0.5 pt + 0.5 pt**]

$$\begin{aligned} K_{ij} &= -GM_m \partial_i \partial_j \left[\left(x_k x^k \right) \right]^{-1/2} \\ &= -GM_m \partial_i \left[- \left(x_k x^k \right)^{-3/2} x_j \right] \\ &= -GM_m \left[-3 \left(x_k x^k \right)^{-5/2} x_i x_j + \left(x_k x^k \right)^{-3/2} \delta_{ij} \right] \end{aligned}$$

(b) The first [article is at $x^i = (0, 0, r) = r\delta^{i3}$. Hence the second particle is at (ξ^1, ξ^2, ξ^3) [**1.5 pt**]

$$\begin{aligned} \frac{d^2\xi^i}{dt^2} &= -K_{ij}\xi^j \\ &= GM_m \left[-3r^{-5}\delta_{i3}\delta_{j3}r^2 + r^{-3}\delta_{ij} \right] \xi^j \\ &= GM_m \left[-\frac{3}{r^3}\delta_{i3}\delta_{j3} + \frac{1}{r^3}\delta_{ij} \right] \xi^j \end{aligned}$$

8. (a) Find dT and dX [**1 pt**]

$$\begin{aligned} dT &= \sinh(gt)dx + gx\cosh(gt)dt \\ dX &= \cosh(gt)dx + gx\sinh(gt)dt \end{aligned}$$

Substituting into ds^2 [**1 pt**]:

$$ds^2 = -(\sinh(gt)dx + gx\cosh(gt)dt)^2 + (\cosh(gt)dx + gx\sinh(gt)dt)^2$$

Cross terms cancel out. Using $\cosh^2(gt) - \sinh^2(gt) = 1$:

$$ds^2 = -g^2x^2dt^2 + dx^2$$

(b) $X^2 - T^2 = x^2$. $X > |T|$.
Given $X > 0$ and $X = x\cosh(gt)$, $x > 0$.

$$x \in (0, \infty), \quad t \in (-\infty, \infty)$$

(c) Non zero components of the metric: $g_{tt} = -g^2x^2$ and $g_{xx} = 1$. Inverse metric components: $g^{tt} = -1/g^2x^2$ and $g^{xx} = 1$. [**0.5 pt**]
3 non-zero Christoffel symbols [**0.5 pt for each correct Christoffel symbol = 1.5 pt**]:

$$\Gamma^x_{tt} = g^2 x$$
$$\Gamma^t_{tx} = \Gamma^t_{xt} = \frac{1}{x}$$